

Name: _____ Maths Class: _____

Sydney Technical High School

2 Unit Mathematics

Year 11

Assessment Task 1 May 2008

General Instructions

- Working time allowed – 70 minutes
- Write using black or blue pen
- Approved calculators may be used
- All necessary working should be shown
- Start each question on a new page
- All questions are of equal value

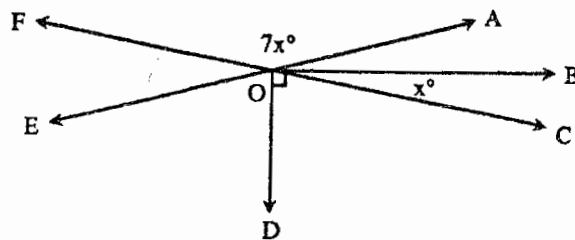
Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Total

Question 1 (7 marks)

- a) Simplify $|-5| - |8|$ 1
- b) Multiply x^3y by $\frac{4y}{x}$ 1
- c) Evaluate $\frac{\sqrt{11.3}}{21.5 \times 0.68}$ to 3 significant figures. 2
- d) The hyperbola $y = \frac{3}{a-x}$ has a vertical asymptote at $x = 1$. What is the value of a . 1
- e) Express the following as fractions in the simplest form
- i) $(3\frac{1}{2})^{-1}$ 1
- ii) $5\frac{1}{3}\%$ 1

Question 2 (7 marks)

- a) Factorise
- i) $9 - 4a^2$ 1
- ii) $3x^2 - 10x + 8$ 2
- b) Solve $2 - 3x < -4$ and graph the solution on a number line 2
- c) AE, CF are straight lines; OD bisects $\angle COE$, $\angle BOD = 90^\circ$



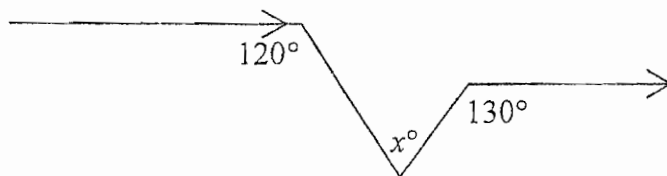
- i) Explain why $\angle EOC = 7x^\circ$ 1
- ii) Find the value of x (no reason) 1

Question 3 (7 marks)

- a) Express $0.3\dot{2}$ as a simple fraction 2
- b) Solve
- i) $\frac{100+p}{p} = \frac{5}{2}$ 2
- ii) $4^{x+1} = 8$ 2
- c) Simplify $(2\sqrt{3})^3$ as a surd 1

Question 4 (7 marks)

- a) By rationalising the denominator express $\frac{3}{3-\sqrt{5}}$ in the form $a + b\sqrt{5}$ 2
- b) Simplify
- i) $\frac{2x}{3} - \frac{x-1}{4}$ 2
- ii) $\sqrt{20} + 3\sqrt{5} + \sqrt{50}$ 2
- c)



Find the value of x (no reasons necessary)

1

Question 5 (7 marks)

a) If $g(x) = x^2 - 6x$

Evaluate

i) $g(-2)$ 1

ii) $g(a - 1)$ 2

b) Use the quadratic formula to solve $2x^2 - 2x - 3 = 0$. Leave the solution in simplified surd form. 2

c) Factorise $m^3 - 8$ 1

d) Write down the domain for the function $y = \sqrt{x + 2}$ 1

Question 6 (7 marks)

a) Sketch the following clearly showing the x and y intercepts.

i) $y = |x + 2|$ 2

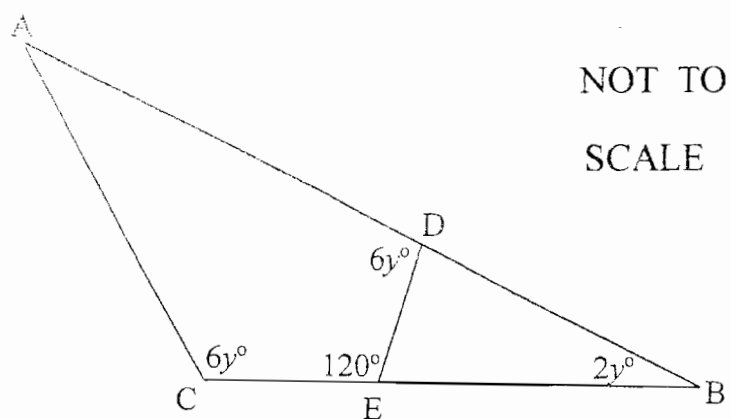
ii) $y = \sqrt{9 - x^2}$ 2

b) If $V = \frac{4}{3}\pi r^3$, find r correct to 2 decimal places when $V = 50$ 2

c) State the range of the function $y = x^2 + 2$ 1

Question 7 (7 marks)

a)



i) Explain why $\angle DEB = 4y^\circ$ 1

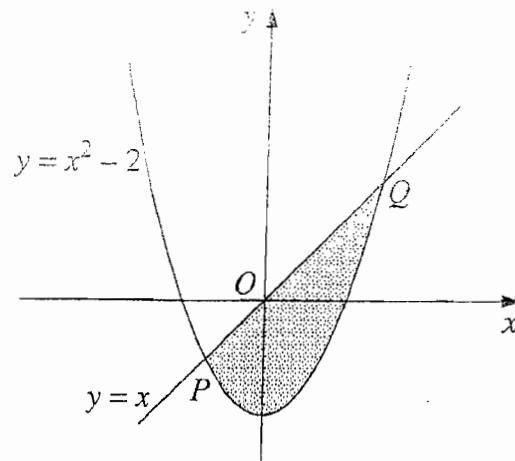
ii) Hence find the size of $\angle CAB$ 2

b) Solve $|2x + 3| = 5$ 2

c) $T = a + (n - 1)d$. Change the subject to n , writing the expression on the right hand side as a single fraction. 2

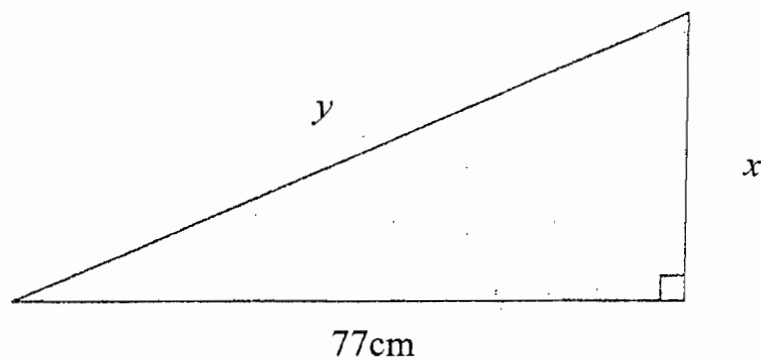
Question 8 (7 marks)

- a) The diagram shows the graphs of $y = x^2 - 2$ and $y = x$.



- i) Find the x values of the points of intersection, P and Q . 2
- ii) Write down the two inequalities which combine to define the shaded region shown. 2

- b) The triangle has a perimeter of 198 cm 3



By forming a pair of simultaneous equations or otherwise find the values of x and y .

Question 1

$$1a) 5 - 8 = -3$$

$$b) \frac{4x^3y^2}{x} = 4x^2y^2$$

$$c) 0.229927\ldots \\ = 0.230$$

$$d) a = 1$$

$$e) i) \frac{2}{7} \\ ii) \frac{4}{75}$$

Question 2

$$a) i) (3 - 2a)(3 + 2a) \\ ii) (3x + 4)(x - 2)$$

$$b) 2 - 3x < -4 \\ -3x < -6 \\ x > 2$$



c) i) vertically opposite angles are equal

$$ii) \frac{9x}{2} = 90^\circ \\ x = 20^\circ$$

Question 3

$$a) \frac{32 - 3}{90} = \frac{29}{90}$$

$$b) i) 200 + 2p = 5p$$

$$3p = 200$$

$$p = 66\frac{2}{3}$$

$$ii) (2^2)^{x+1} = 2^3$$

$$2^{2x+2} = 2^3$$

$$2x + 2 = 3$$

$$x = \frac{1}{2}$$

$$c) (2\sqrt{3})^3 = 8\sqrt{27} \\ = 24\sqrt{3}$$

Question 4

$$a) \frac{3}{3 - \sqrt{5}} \times \frac{3 + \sqrt{5}}{3 + \sqrt{5}} = \frac{9 + 3\sqrt{5}}{4}$$

$$b) i) \frac{8x - 3(x - 1)}{12}$$

$$= \frac{5x + 3}{12}$$

$$ii) 2\sqrt{5} + 3\sqrt{5} + 5\sqrt{2}$$

$$= 5\sqrt{5} + 5\sqrt{2}$$

$$c) x = 70^\circ$$

Question 5

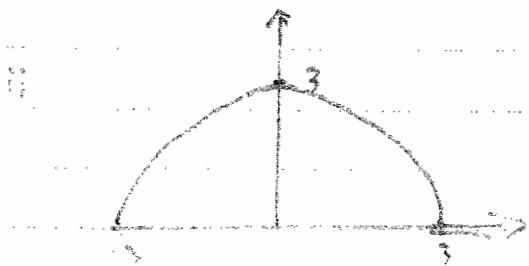
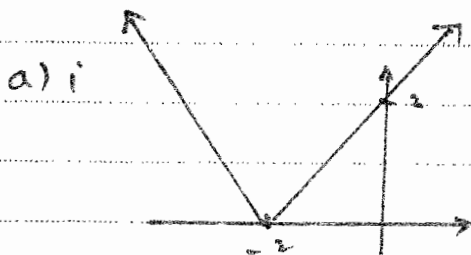
$$a) \log(-2) = 4 + 12 \\ = 16$$

$$ii) \log(a-1) = (a-1)^2 - 6(a-1) \\ = a^2 - 2a + 1 - 6a + 6 \\ = a^2 - 8a + 7$$

$$b) x = \frac{2 \pm \sqrt{4 + 24}}{4} \\ = \frac{2 \pm \sqrt{28}}{4} \\ = \frac{2 \pm 2\sqrt{7}}{4} \\ = \frac{1 \pm \sqrt{7}}{2}$$

$$c) m^3 - 8 = (m-2)(m^2 + 2m + 4)$$

$$d) x > -2$$

Question 6

$$b) 50 = \frac{4}{3} \pi r^3 \\ r = \sqrt[3]{\frac{3}{4} \cdot 50}$$

$$c) y > 2$$

Question 7

a) i; exterior angle of triangle (equals sum of interior opposite angles)

$$ii) 4y + 120^\circ = 180^\circ \\ y = 15^\circ$$

$\therefore \angle CAB + 90^\circ + 30^\circ = 180^\circ$
angle sum of triangle.

$$\angle CAB = 60^\circ$$

$$b) 2x + 3 = 5 \\ 2x = 2 \\ x = 1$$

or.

$$2x + 3 = -5$$

$$2x = -8$$

$$x = -4$$

$$c) T = a + (n-1)d$$

$$T - a = (n-1)d$$

$$\frac{T-a}{d} = n-1$$

$$n = \frac{T-a}{d} + 1$$

$$= \frac{T-a}{d} + d$$

Question 8

a) i) $x^2 - 2 = x$

$$x^2 - x - 2 = 0$$

$$(x - 2)(x + 1) = 0$$

$$x = 2 \text{ or } -1$$

For $y = x$

ii) at $(0, -1)$ $-1 \leq 0$

$$\therefore y \leq x$$

For $y = x^2 - 2$

at $(0, -1)$ $-1 \geq -2$

$$\therefore y > x^2 - 2$$

b)

$$x + y + 77 = 198$$

$$x + y = 121 \quad (1)$$

$$x^2 + 77^2 = y^2$$

$$y^2 - x^2 = 5929$$

$$(y - x)(y + x) = 5929$$

$$\therefore y - x = 49 \quad (2)$$

 $(1) + (2)$

$$2y = 170$$

$$y = 85$$

$$\therefore x = 36$$

$$\therefore x = 36, y = 85$$